

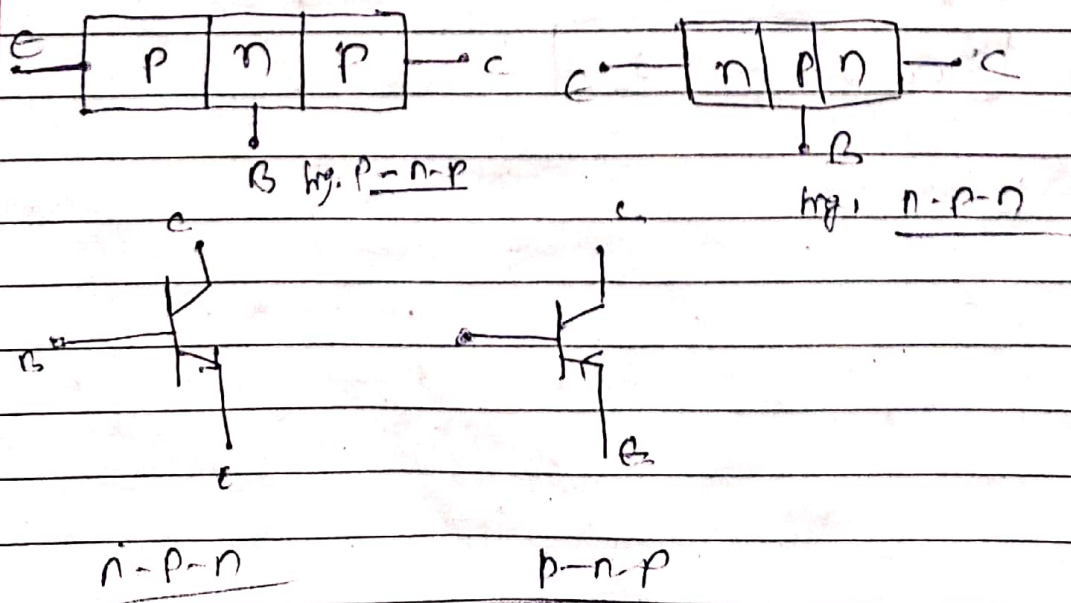
Bipolar Junction Transistor

A transistor consists of two p-n junction. It is formed by sandwiching either p-type or n-type semiconductor betⁿ a pair of opposite types.

There are two types of transistor may be,

- i) n-p-n transistor
- ii) p-n-p transistor.

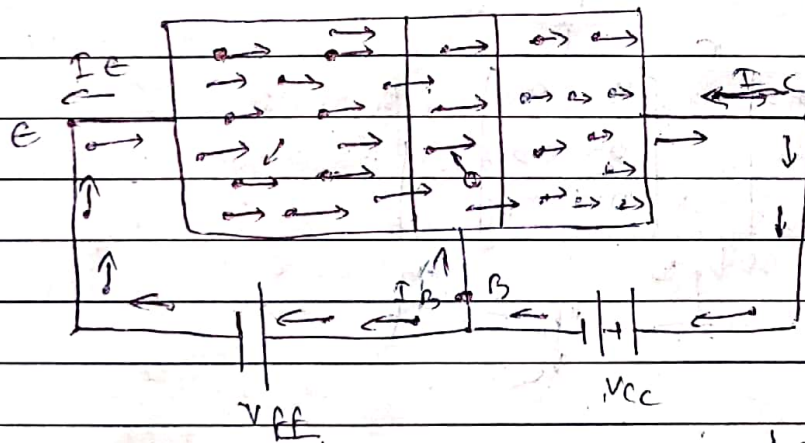
↳ n-p-n transistor is composed of two n-type semiconductor separated by a thin section of p-type and p-n-p transistor is formed by two p-type semiconductor separated by thin section of n-type as shown in fig.



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Transistor consists of three terminals namely emitter (E), Base (B), and collector (C). The emitter region is highly doped, collector region is moderately doped and base region is lightly doped. The emitter base junction is forward biased whereas the collector base junction is reverse biased.

Working of n-p-n transistor:



Emitter base Junction: is forward biased and collector base junction is reverse biased. As shown in fig above. The forward biased causes the electrons n-type emitter to flow towards the base. This constitutes the emitter current I_E . As these electrons flow through p-type base they tend to combine with holes. As the base is lightly doped and very thin, only few electrons (less than 5%) combine with holes

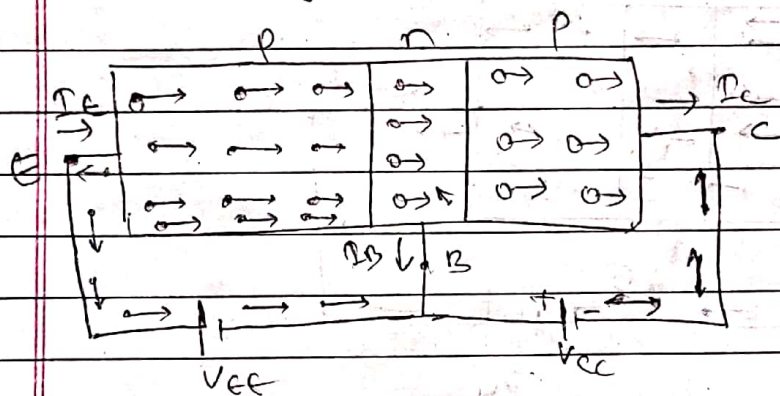
do constitute base current I_B .

The remainder (more than 95%) crossover into the collector region.

do constitute collector current I_C .

In this way almost the entire emitter current flows in the collector circuit. The emitter current is the sum of collector current and base current.

Working of p-n-p transistor:

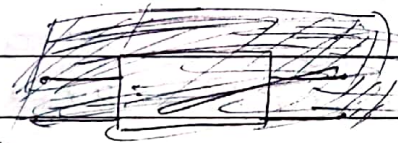
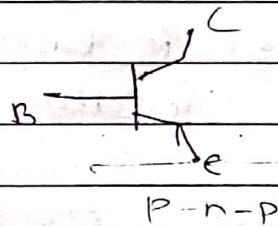
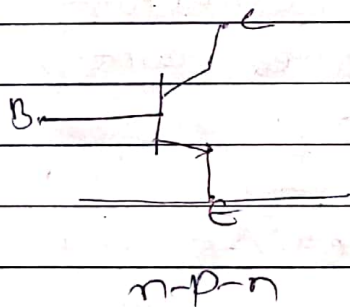


The forward bias meter causes the holes in p-region to flow toward the base. This constitutes the emitter current I_E . At these holes, they go through base, n-type base, tends to combine to electron. As the base is lightly doped and very thin only few holes (less than 5%), combines with electrons to constitute base current I_B . The remainder (more than 95%)

flow over into the collector region to constitute collector current I_c . In this way almost the entire emitter current flows in collector ckt. The entire emitter current is the sum of collector and base current.

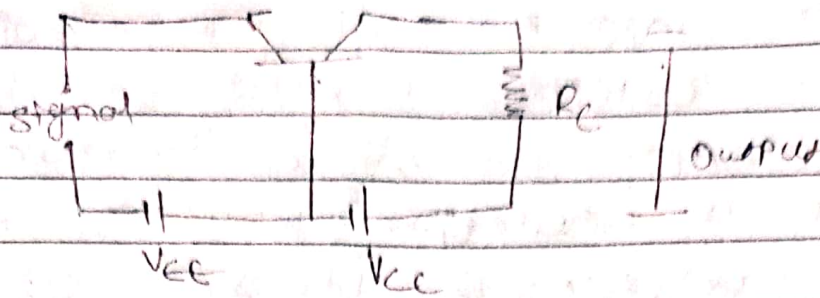
It may be noted that current conduction within p-n-p transistor is by holes. However in external connecting wire the current conduction is due to electrons.

Transistor Connection:



- 1) common base connection
- 2) common emitter connection
- 3) " collector "

1) common base connection: In this arrangement input is applied betⁿ emitter and base and output is taken from collector and base as shown in fig.



Here, base is common to both input and output ckt. Hence, this arrangement is known as common base connection / configuration.

The ratio of change in collector current to the change in emitter current at constant collector base voltage is known as current amplification factor.

$$\alpha = \frac{\Delta I_C}{\Delta I_E} \text{ at const } V_{CB}$$

The value of current amplification factor is less than 1.

Expression for collector current:

The total collector current consists of:
 (i) Part of emitter current which reaches the collector terminal, i.e. ΔI_C

(ii) The leakage current ($I_{leakage}$), is due to the movement of minority carrier.

across the base collector Junction.
This is generally much smaller than
(αI_E). Therefore,

$$\text{Collector current } (I_C) = I_E + I_{\text{leakage}}$$

If $I_E = 0$, that is emitter ckt is open
a small leakage current still flows in
the collector ckt this leakage current
is ~~very~~ abbreviated as I_{CBO} .

$$\therefore I_C = \alpha I_E + I_{CBO}$$

$$I_C = \alpha (I_C + I_B) + I_{CBO}$$

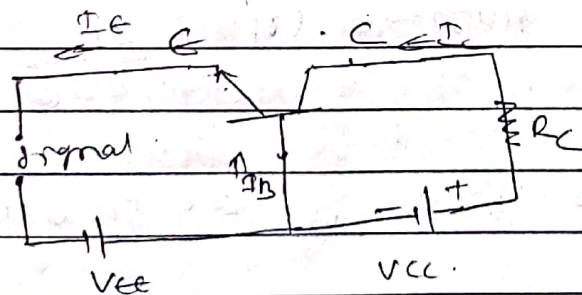
$$I_C = \alpha I_B + \alpha I_C + I_{CBO}$$

$$I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$I_C = \frac{\alpha I_B + I_{CBO}}{(1 - \alpha)} \quad \text{--- (11)}$$

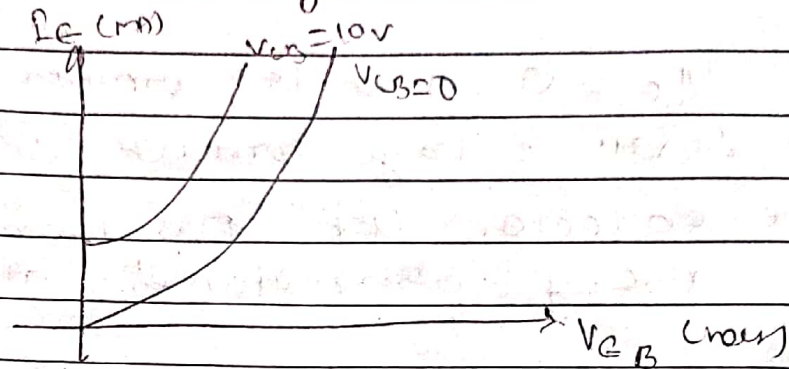
(imp)

Input and Output characteristics:



It is the curve between emitter current
 I_E and emitter base voltage V_{EB} at const.
collector base voltage V_{CB} . The emitter
current is generally taken along y-axis
and emitter base voltage along x-axis.

The emitter current (I_E) increases rapidly with small increase in emitter base voltage. The emitter current is almost independent of collector base voltage;



Input Characteristics

The ratio of change in emitter base voltage to resulting change in emitter current at constant collector base voltage is known as Input resistance.

$$\text{i.e., Input resistance } (r_i) = \frac{\Delta V_{BE}}{\Delta I_E}$$

The input resistance is very small.

CB connection

Output characteristics:

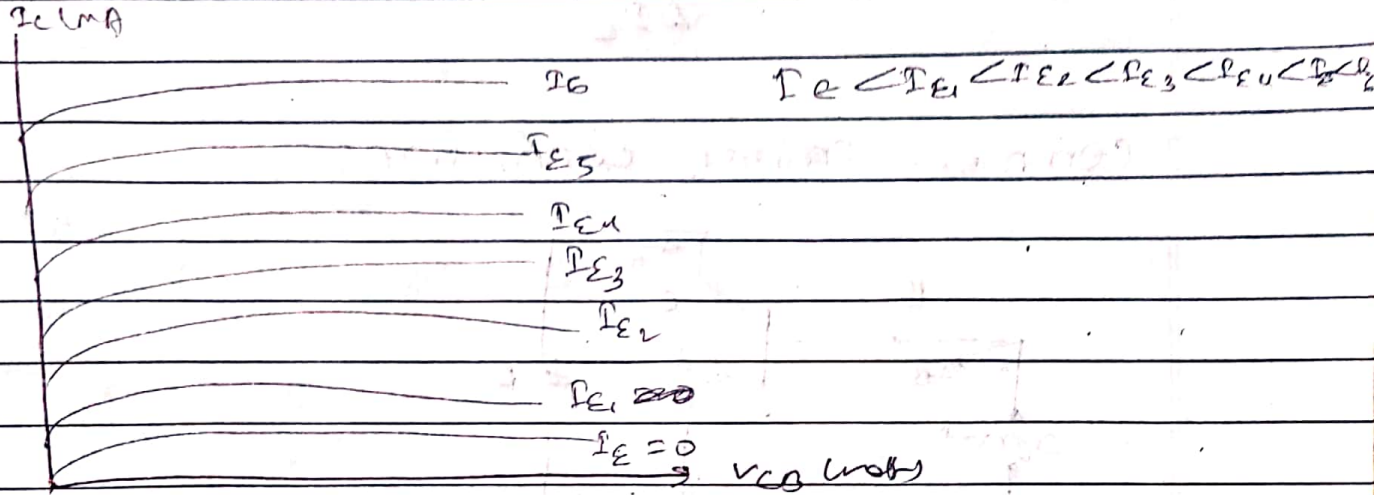


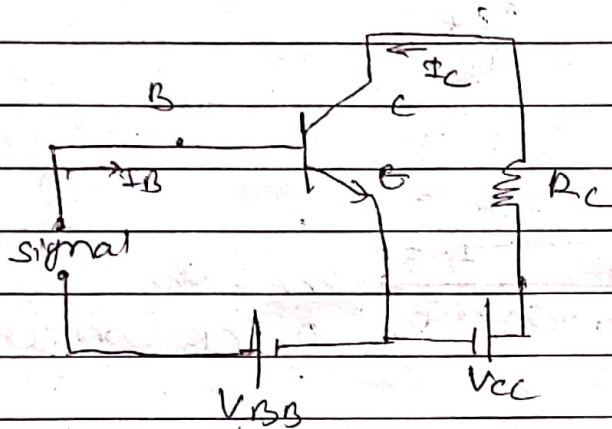
Fig. Output characteristics of CB connection curve been

It is the collector current I_C and emitter current. Generally, collector current is taken along y-axis and collector base voltage is taken along x-axis. The collector current (I_C) varies with V_{CB} only at very low voltage (less than ϕ). The transistor is never operated in this region, when value of V_{CB} is increased, the collector current becomes const. as indicated by straight line. The transistor is operated in this region. A very large change in collector base voltage produces only a small change in collector current. This means that the output resistance is very high. The output resistance is given by the ratio of change in collector base voltage to the resulting change

in collector current at constant emitter current.

$$i.e. r_o = \frac{\Delta V_{CB}}{\Delta I_C} \text{ at const } I_E$$

Common Emitter connection?



In this arrangement input is applied betⁿ base and emitter and output is taken from collector and emitter. Here, emitter of transistor is common to both input and output ckt. Therefore this arrangement is known as Common emitter configuration.

Current amplification factor (β):

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

→ It is the ratio of change in collector current to the change in base current.

Relationship between α and β :

we have,

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$\& \beta = \frac{\Delta I_C}{\Delta I_B}$$

$$\text{As, } I_E = I_B + I_C$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

Dividing both side by ΔI_C ,

$$\frac{\Delta I_E}{\Delta I_C} = \frac{\Delta I_B}{\Delta I_C} + 1$$

$$\Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\Rightarrow \frac{1}{\beta} = \frac{1}{\alpha} - 1 = \frac{1-\alpha}{\alpha}$$

$$\Rightarrow \beta = \frac{\alpha}{1-\alpha} \quad \text{--- } \textcircled{*}$$

Expression of collector current:

$$I_C = \alpha I_E + I_{CBO}$$

also,

$$I_E = I_C + I_B$$

$$\therefore I_C = \alpha (I_B + I_C) + I_{CBO}$$

$$= \alpha I_B + \alpha I_C + I_{CBO}$$

$$\therefore I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$\therefore I_C = \frac{\alpha}{(1-\alpha)} I_B + \frac{I_{CBO}}{(1-\alpha)}$$

If $I_B = 0$, the I_C is attributed as I_{CEO}

$$\therefore I_{CEO} = \frac{I_{CBO}}{(1-\alpha)}$$

$$I_C = \frac{\alpha}{(1-\alpha)} I_B + I_{CEO}$$

$$I_C = \beta I_B + I_{CEO} \quad \text{--- (xx)}$$

Input characteristics:

It is a curve betⁿ base current I_B and base emitter voltage (V_{BE}) at constant collector emitter voltage (V_{CE}). Generally, base current is taken along Y-axis and base emitter voltage is taken along X-axis. The input characteristics are

- 1) Similar to that of forward biased diode curve as compared to CB (common base arrangement) I_B increases less rapidly with V_{BE} . Therefore, Input resistance of CE arrangement is higher than that of CB ckt/arrangement. Input resistance is the ratio of change in base emitter voltage to the resulting change in base current at V_{CE} . i.e.

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B} \text{ at const } V_{CE}$$

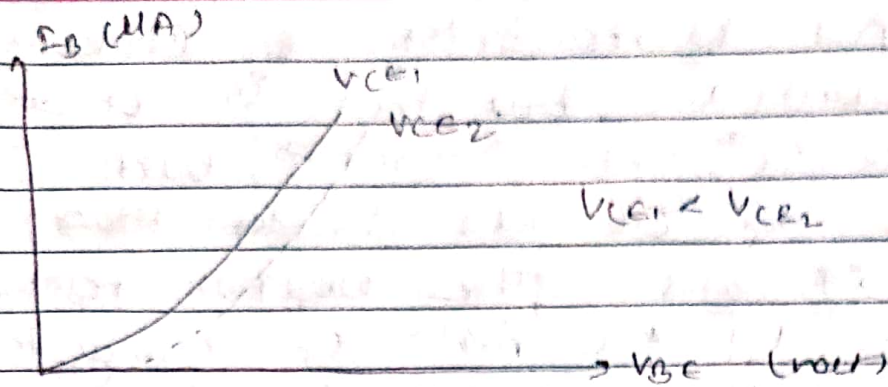
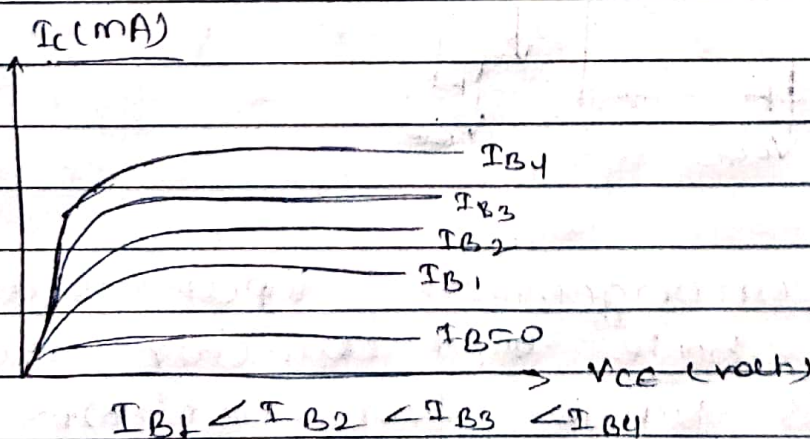


Fig. Input characteristics
of common emitter connection.

Output characteristics:

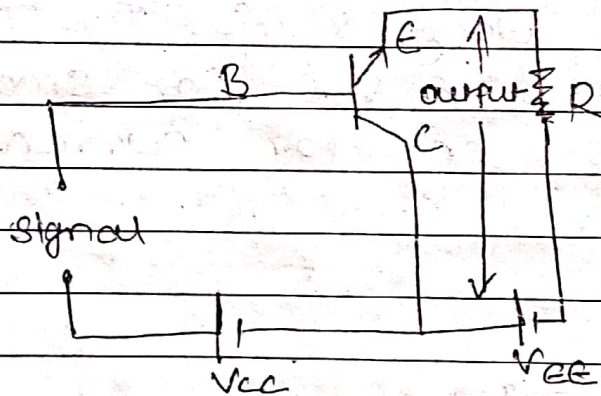
→ It is the curve between collector emitter voltage and collector current at constant base current. Generally, collector current is taken along y-axis and collector emitter voltage is taken along x-axis. The collector current I_C varies with V_{CE} for small value of V_{CE} . After this collector current becomes almost constant.



The output characteristics of CB ckt are horizontal but there is noticeable slope for CE ckt. Therefore, output resistance of CE ckt is less than that of CB ckt. The output resistance is given by the ratio of change in collector emitter voltage to the resulting change in collector current at constant I_B .

$$\text{i.e. } r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at const } I_B.$$

Common collector connection :



In this arrangement, input is applied between base and collector and output is taken from emitter and collector. Here, collector is common to both input and output ckt. Therefore, this ckt is known as common collector

Connection:

Current Amplification factor (γ):
 It is the ratio of ^{change in} emitter current to change in base current.

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

Relationship betⁿ β and γ :

we know,

$$\beta = \frac{\Delta I_C}{\Delta I_B} \text{ and } \gamma = \frac{\Delta I_E}{\Delta I_B}$$

Since,

$$I_E = I_B + I_C$$

$$\Delta I_E = \Delta I_B + \Delta I_C$$

Dividing both sides by ΔI_B ,

$$\frac{\Delta I_E}{\Delta I_B} = 1 + \frac{\Delta I_C}{\Delta I_B}$$

$$\gamma = 1 + \beta$$

$$\boxed{\beta = \gamma - 1}$$

Relationship betⁿ α , β , γ :

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\beta = \gamma - 1$$

$$\gamma - 1 = \frac{\alpha}{1-\alpha}$$

$$\gamma = \frac{\alpha}{1-\alpha} + 1 = \frac{\alpha + 1 - \alpha}{1-\alpha}$$

$$\Rightarrow \gamma = \frac{1}{1-\alpha}$$

Comparison betⁿ CB, CE and CC arrangement:

Parameters	CB	CE	CC
Input resistance	low	low but higher than CB	very high
Output resistance	very high	high	low
Voltage gain	low	medium	less than 1
Current gain	less than unity	high	medium
Application	High freq. Application	Audio freq. Application	Impedance matching.

Why common emitter arrangement is used in most of transistor application out of three connection common emitter ckt is most efficient. The main reason for the wide spread use of common ckt arrangement are;

- I) High current gain
- II) High voltage and power gain
- III) moderate output to input impedance ratio.

Transistor as an amplifier!

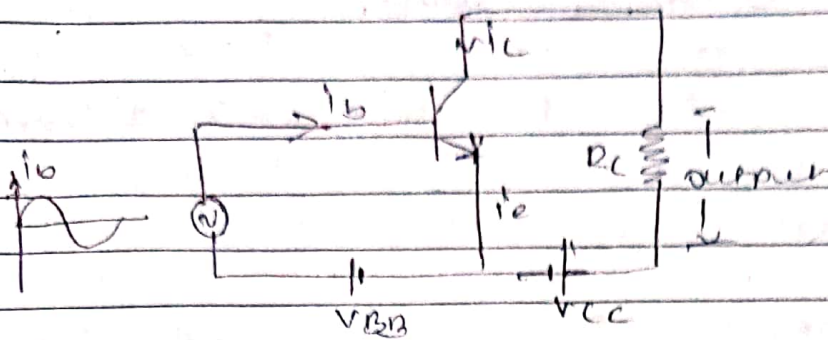


Fig. Input signal

Common emitter n-p-n transistor amplifier circuit is shown in fig above. Here, a battery V_{BB} is connected in input circuit in addition to the signal voltage. This dc voltage is known as bias voltage and its magnitude is such that, it always keeps the emitter base junction forward biased regardless of the polarity of the signal.

During positive half cycle of input signal, the forward bias across emitter base junction is increased hence more electrons flow from emitter to collector through base. This causes an increase in collector current, the increased collector current produces a greater voltage drop across collector load resistance R_C . However during the -ve half cycle of the signal the forward bias across emitter base junction is decreased, therefore collector current decreases. This results in the decrease in output voltage (in oppo

same direction). Hence, amplified output is obtained across the load.

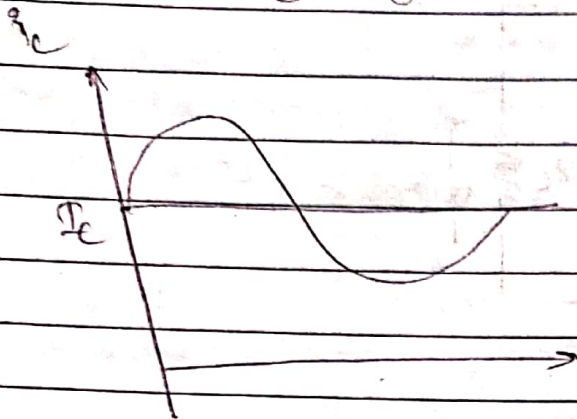
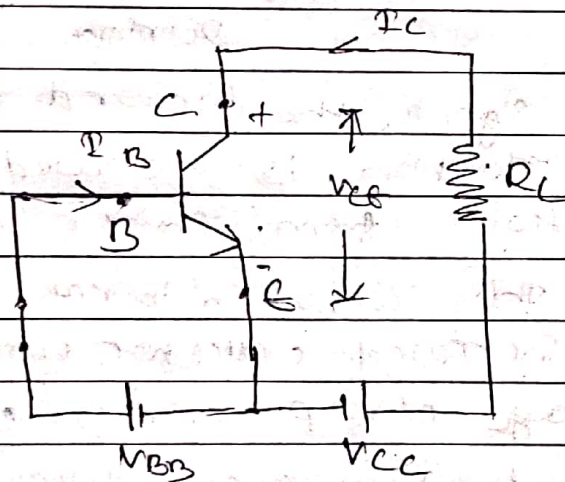


Fig. Output signal

Transistor load line analysis or load line:

Let us consider a common emitter amplifier ckt as shown in fig.



Applying KVL at output side,

$$V_{CC} = I_C R_C + V_{CE} \quad \text{--- (1)}$$

$$V_{CE} = V_{CC} - I_C R_C \quad \text{--- (1)}$$

Equation ① gives the value of collector emitter voltage V_{CE} at any time as V_{CC} and R_C are fixed. Therefore eqn ① is first degree equation and can be represented by straight line in the output characteristics. This line is known as DC load line and determines the locus of $V_{CE} - I_{CE}$ points for the given value of R_C . To draw DC load line we need two points of the straight line, these two points can be found as;

put $I_C = 0$ in eqn ①,

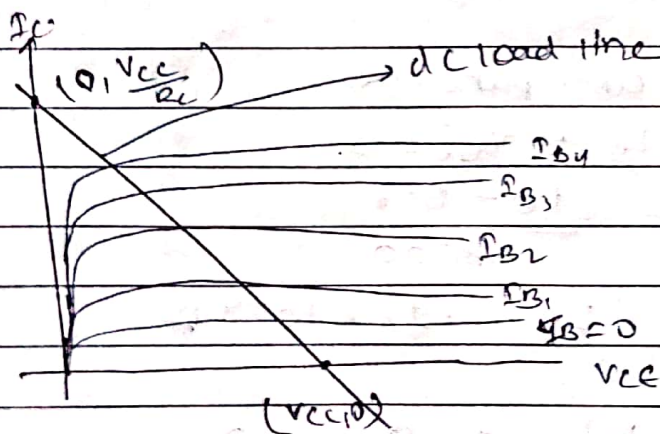
$$V_{CE} = V_{CC}$$

∴ One point of st line is $(V_{CC}, 0)$

put $V_{CE} = 0$ in eqn ①,

$$I_C = \frac{V_{CC}}{R_C}$$

Another point of st line is $(0, \frac{V_{CC}}{R_C})$

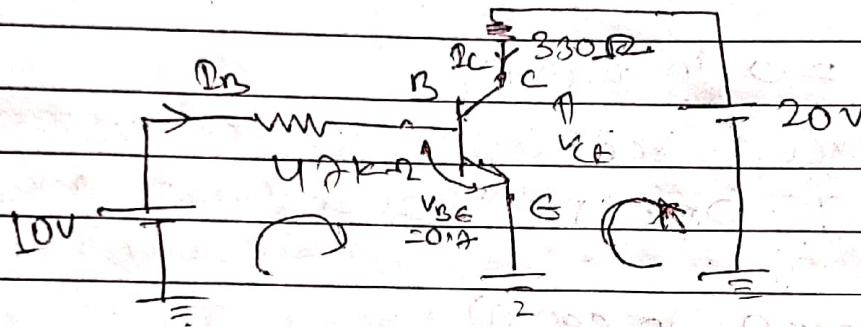


* Operating point (Q-point):

→ zero-signal value of I_c and V_{ce} is known as operating point, is also called quiescent point (Q-point). This point is called operating point because variation of I_{CQ} and V_{CE} takes place.

Q. Determine the Q-point of the transistor circuit shown in fig. below. Also draw the dc load line. Given, $\beta = 200$.

→



$$V_{BE} = 0.7 \quad (\text{if not given silicon: } 0.7)$$

$$\beta = 200$$

Applying KVL at input side

$$10 - 47000 I_B - V_{BE} = 0$$

$$\therefore I_B = \frac{10 - 0.7}{47000}$$

$$= 1.97 \times 10^{-4} \text{ A}$$

$$\text{Since, } I_C = \beta I_B = 0.394 \text{ A}$$

Applying KVL at output side, (1)

$$20 - 330 I_c - V_{CE} = 0$$

$$\therefore V_{CE} = 20 - 330 \times 0.0394$$

$$= 7 \text{ V}$$

operating point $(V_{CE}, I_c) = (7, 39.4 \text{ mA})$

for dc load line,

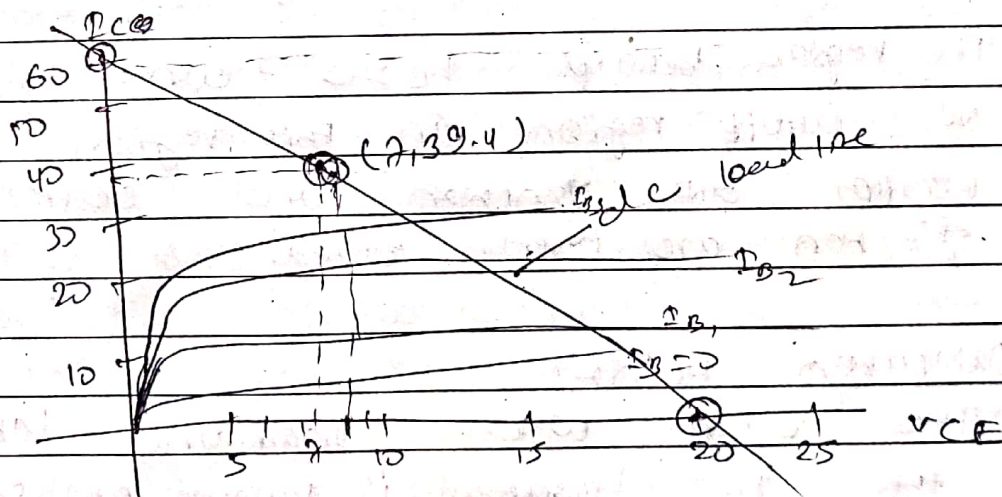
put $I_c = 0$ in eqn (1).

$$20 - 0 - V_{CE} = 0$$

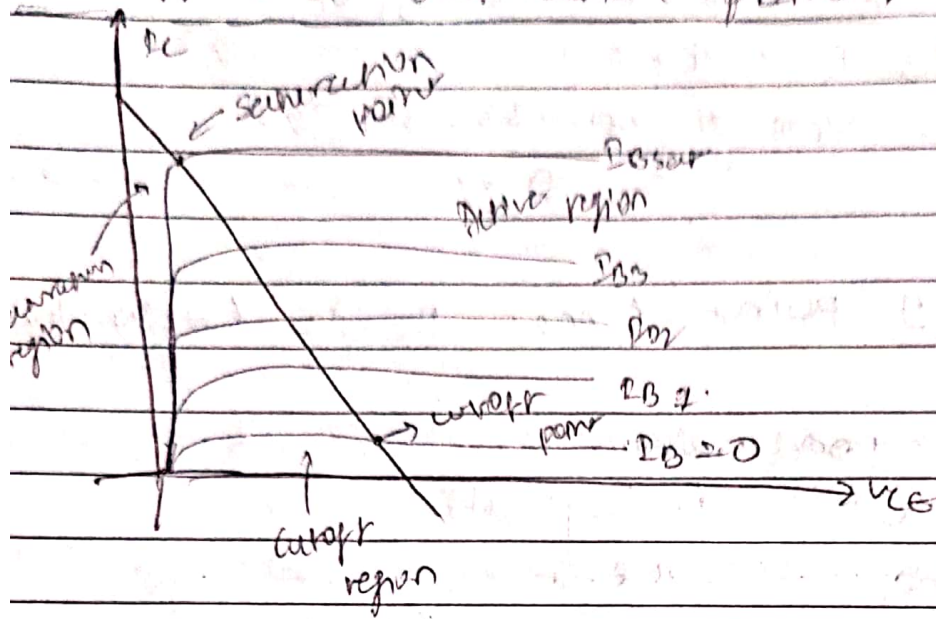
$$V_{CE} = 20 \text{ V} \quad (20, 0)$$

put $V_{CE} = 0$, in eqn (1)

$$I_c = \frac{20}{330} = 0.061 = 61 \text{ mA} \quad (0, 61)$$



Cutoff and saturation points:



cutoff point:

The point where load line intersects $I_B = 0$ curve is known as cutoff point.

The region below $I_B = 0$ curve is known as cutoff region. In this region both emitter base junction and collector base junction are reverse biased.

saturation point:

The point where load line intersects the $I_B = I_{Bsat}$ curve is known as saturation point.

In the saturation region both emitter base junction and collector base junction are forward biased.

Active region:

The region between cutoff and saturation is known as active region. In this region collector base junction remains reverse biased while emitter base junction remains forward biased. The transistor will function normally in this region.

Faithful amplification:

The process of rising strength of weak signal without any change in its general shape is known as faithful amplification.

Transistor Biasing:

The proper flow of zero signal collector current and maintenance of proper emitter collector voltage during the passage of signal is known as transistor biasing.

The basic purpose of transistor biasing is to keep the base emitter junction properly forward biased and collector base junction properly reverse biased during the application of signal. This can be achieved with bias battery or associating a DC with a transistor. The DC which provides

transistor biasing is known as biasing DC. The biasing DC must have the follow

ing condn:

- i) The Q-point must lie in the middle of the active region. It must not move either to cutoff or saturation when input signal is applied.
- ii) The collector current should not vary with temp. variation.
- iii) The Q-point be independent of ' β ' i.e. it should not change if transistor is replaced by the new transistor having similar characteristics.

Stabilization:

The process of making operating point independent of temp. change or variation in transistor parameter is known as stabilization.

Stabilization of operating point is necessary due to,

- i) temp. dependence of I_c
- ii) individual variation
- iii) thermal runaway.

Stability factor (S):

The rate of change of collector current I_C with respect to collector leakage current (I_{CBO}) at const. β and I_B is called stability factor.

$$I.C.S = \frac{dI_C}{dI_{CBO}} \text{ at const } I_B \text{ \& } \beta.$$

we have

$$\text{collector current } (I_C) = \beta I_B + (1+\beta) I_{CBO} \quad \text{--- (1)}$$

diff. eqn wrt to I_C

$$\frac{dI_C}{dI_C} = \beta \frac{dI_B}{dI_C} + (1+\beta) \frac{dI_{CBO}}{dI_C}$$

$$1 = \beta \cdot \frac{dI_B}{dI_C} + (1+\beta) \cdot \frac{1}{S}$$

$$S = \frac{(1+\beta)}{(1 - \beta \cdot \frac{dI_B}{dI_C})} \quad \text{--- (II)}$$

→ Higher value of stability factor results less indicates less stability of operating point. In order to achieve greater thermal stability, it is desirable to have no stability factor as possible.

Ques. # Method of transistor biasing :-

The most commonly used methods of transistor biasing are :

- I) Base resistor method (fixed bias ckt)
- II) Biasing with collector feedback resistor (collector to base bias ckt)
- III) Voltage divider biasing

I) Base resistor method :

→ In this method, a high resistance R_B is connected between base and positive end of supply for n-p-n transistor. Here, required zero signal base current is provided by V_{CC} and it flows through R_B .

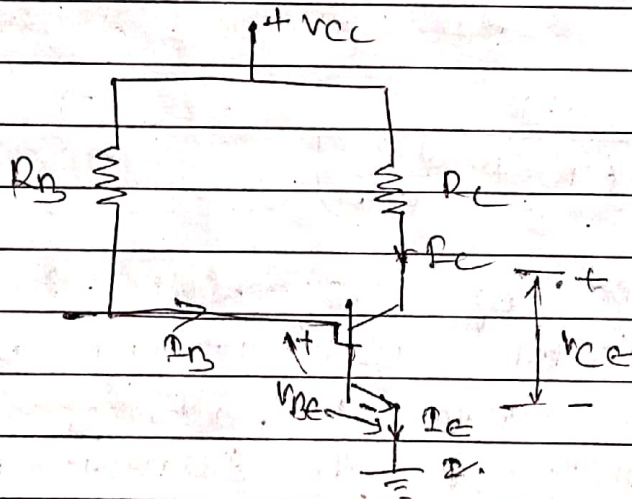


Fig. 1

Applying KVL at input side,

$$V_{CC} - I_B R_B - V_{BE} = 0 \quad \text{--- (1)}$$

Applying KVL at Output side,
 $V_{CC} + I_C R_C - V_{CE} = 0 \quad \text{--- (1)}$

for load line, put $I_C = 0$
 $V_{CE} = V_{CC}$

Again, put $V_{CE} = 0$,

$$I_C = \frac{V_{CC}}{R_C}$$

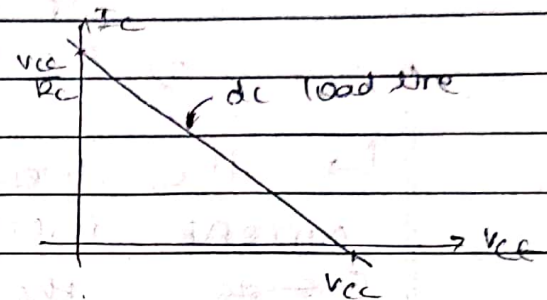


Fig. 2.

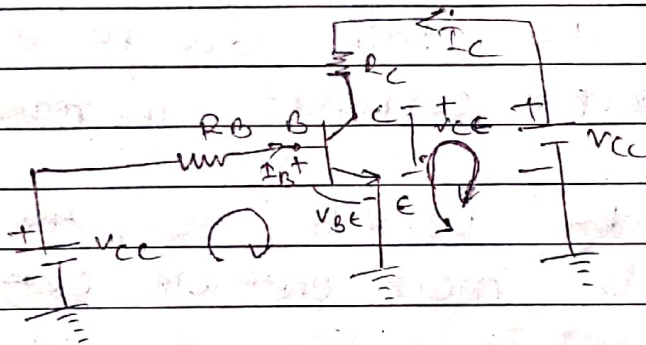


Fig. 3. similar as above (Fig 1)

$$S = \frac{I + \beta}{1 - \beta \frac{dV_{BE}}{dI_C}}$$

Now, diff eqn (1) wrt I_C

$$\frac{dV_{CC}}{dI_C} - R_C \frac{dI_C}{dI_C} - \frac{dV_{CE}}{dI_C} = 0$$

$$0 - R_C \frac{dI_C}{dI_C} - 0 = 0$$

$$\Rightarrow \frac{dI_C}{dI_C} = 0$$

$$\text{Stability factor } (S) = \frac{1+\beta}{1 - \beta \frac{dI_B}{dI_C}}$$

$$\Rightarrow S = 1+\beta$$

if $\beta = 100$,

$$\Rightarrow S = 101$$

↳ This means that if leakage saturation current increased by $1 \mu A$ due to temp. change, the collector current increases by $(1+\beta)$ times.

i.e. I_C increases by $101 \mu A$, so, the fixed bias ckt is most unstable ckt.

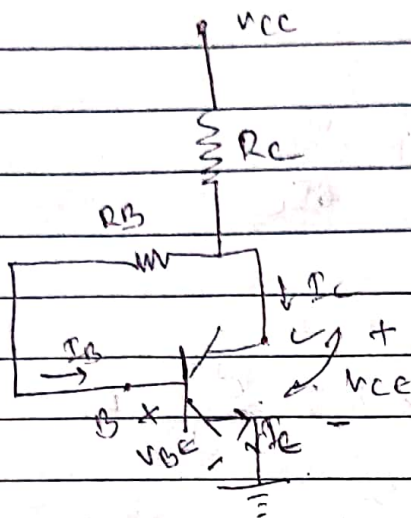
Advantage:

- I) Biasing ckt is very simple
- II) Biasing condn can easily set and calculation is simple
- III) There is no loading of the source by the biasing ckt.

Disadvantage:

- I) This method provides poor stabilization
- II) Stability factor is very high, therefore, there is strong chance of thermal runaway.

Collector to Base resistor method + bias circuit:



In this method, One end of R_B is connected to the base and other end is connected to other end as shown in fig. above. Here, the required zero signal base current is determined by collector base ~~current~~ voltage. The collector base voltage forward biases the base emitter junction and hence, base current I_B flows through R_B .

Applying KVL in input side,

$$V_{CC} - (I_B + I_C)R_C - I_B R_B - V_{BE} = 0$$

$$\text{or, } V_{CC} - I_B R_C - I_C R_C - I_B R_B - V_{BE} = 0$$

$$\text{or, } V_{CC} - I_B (R_C + R_B) - V_{BE} - I_C R_C = 0 \quad \text{--- (1)}$$

KVL in output side,

$$V_{CC} - (I_B + I_C)R_C - V_{CE} = 0 \quad \text{--- (2)}$$

Stability factor (S):

we know,

$$S = \frac{\beta + 1}{1 - \beta \frac{dI_B}{dI_C}}$$

Diff eqn ① w.r.t. I_C ,

$$\frac{dV_{CE}}{dI_C} - \frac{dI_B}{dI_C} (R_C + R_B) - \frac{dR_C}{dI_C} R_C - \frac{dV_{BE}}{dI_C} = 0$$

$$\Rightarrow 0 - \frac{dI_B}{dI_C} (R_C + R_B) - R_C = 0 \quad \text{--- (2)}$$

$$\Rightarrow \frac{dI_B}{dI_C} = - \frac{R_C}{R_C + R_B}$$

$$S = \frac{\beta + 1}{1 + \beta \frac{R_C}{R_C + R_B}}$$

$$\text{Let } \beta = 100, \quad R_C = 2k, \quad R_B = 100k$$

$$S = \frac{100 + 1}{1 + 100 \times \frac{2}{2 + 100}}$$

$$= 34.11$$

Stability factor of collector to load bias ckt. is less than that of fixed bias ckt. so it is more stable than fixed bias ckt.

↳ This simple method as it requires only one resistor (R_B).

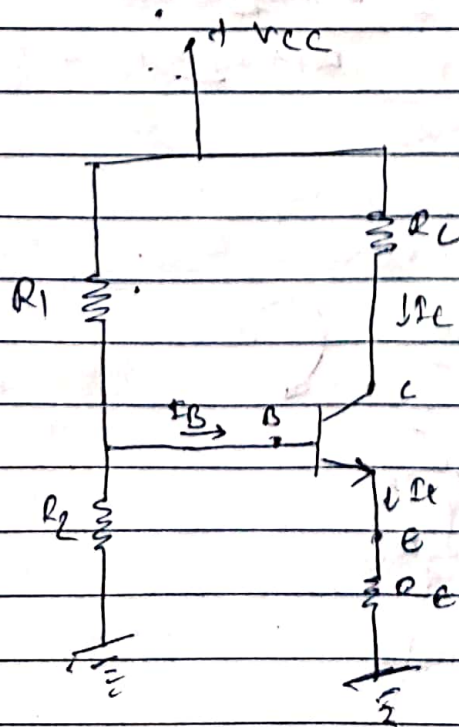
↳ The ckt provides some stabilization of the operating point.

Disadvantage:

↳ This ckt doesn't provide good stabilization becoz stability factor is still fairly high.

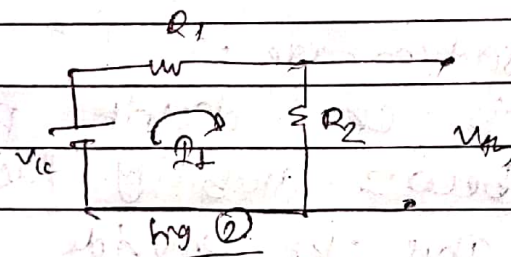
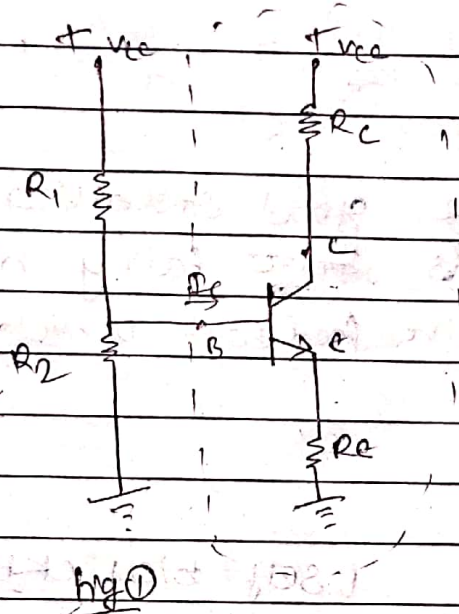
↳ This ckt provides negative feedback which reduces the gain of amplifier.

Voltage divider biasing: (Self bias ckt):

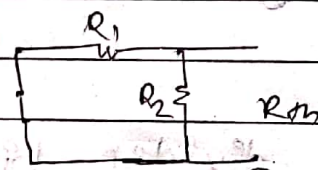


This is the most widely used method of providing biasing. In this method, two resistors R_1 & R_2 are connected across V_{CC} and this provides biasing. The emitter resistance R_E provides stabilization, now in order to solve the ckt, convert the input ckt into its equivalent Thevenin's ckt,

converting input side into Thevenin's eq. ckt,

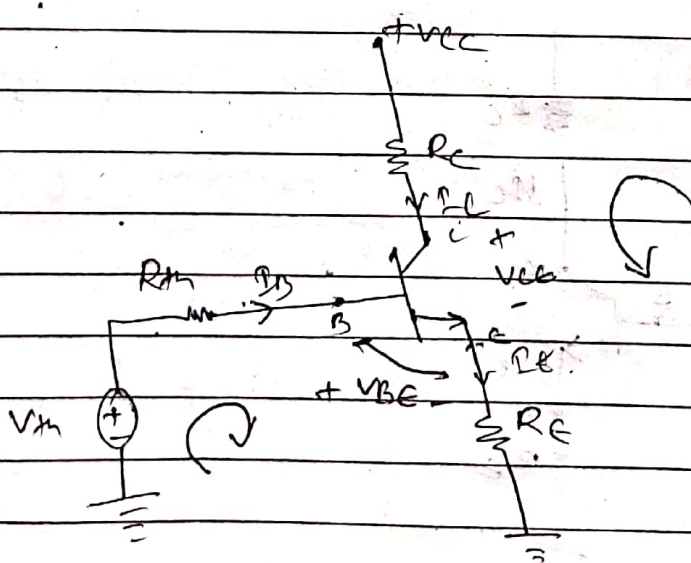


$$\Rightarrow V_{TH} = I_B \times R_2 = \frac{V_{CC} \times R_2}{R_1 + R_2}$$



$$\Rightarrow R_{TH} = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$I_B = \frac{V_{CC}}{R_1 + R_2}$$



Apply KVL input side,

$$V_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$$

$$\text{or } V_{th} - I_B R_{th} - V_{BE} - (I_B + I_C) R_E = 0$$

$$\text{or } V_{th} - I_B (R_{th} + R_E) - V_{BE} - I_C R_E = 0 \quad \text{--- (1)}$$

Apply KVL output side,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\text{Stability factor (S)} = \beta + 1$$

$$1 - \beta \frac{dI_B}{dI_C}$$

diff. eqn (1) wr to I_C ,

$$\frac{dV_{th}}{dI_C} - (R_{th} + R_E) \frac{dI_B}{dI_C} - \frac{d(V_{BE})}{dI_C} - R_E \frac{dI_C}{dI_C} = 0$$

$$\text{or } 0 - (R_{th} + R_E) \frac{dI_B}{dI_C} - 0 - R_E = 0$$

$$\text{or } \frac{dI_B}{dI_C} = - \frac{R_E}{(R_{th} + R_E)}$$

$$S = \beta + 1$$

$$1 + \beta \frac{R_E}{(R_{th} + R_E)}$$

$$(R_{th} + R_E)$$

$$\text{Let } \beta = 100, R_E = 1k$$

$$R_1 = 56k, R_2 = 12k$$

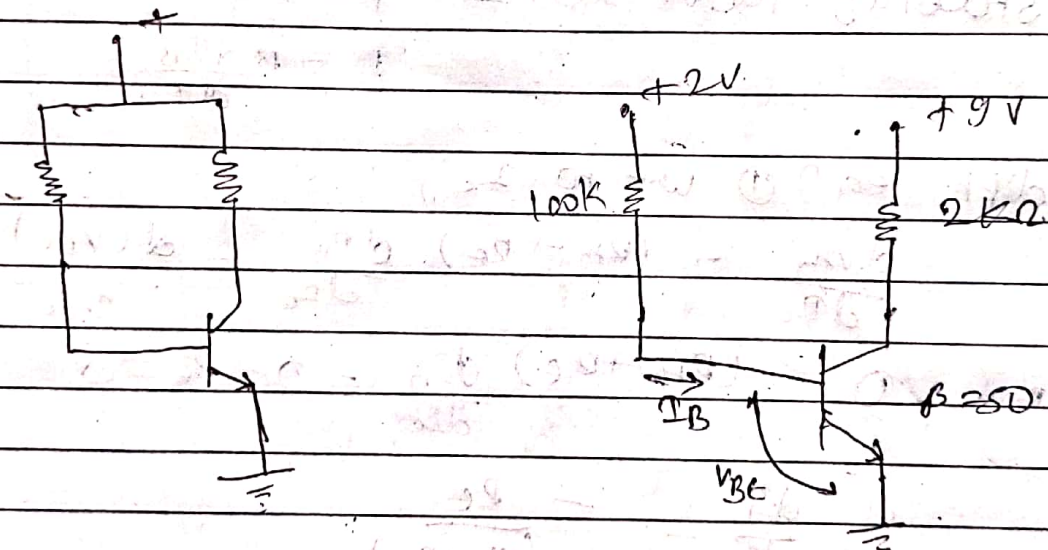
$$R_{th} = 9.88k$$

$$S = 9.91$$

Self bias or voltage divider ckt is the most stable ckt compare to fixed bias and collector base bias ckt.

Ans: +

Q. Find the operating point of the given ckt. Also draw the dc load line.



2)

Applying KVL at input side,

$$2 - 100I_B - V_{BE} = 0$$

$$\therefore I_B = \frac{2 - V_{BE}}{100}$$

$$= \frac{2 - 0.7}{100}$$

$$= 0.013 \text{ mA}$$

As,

$$I_C = \beta I_B$$

$$\text{As, } I_C = \beta I_B$$

$$= 50 \times 0.013$$

$$= 0.65 \text{ mA}$$

Applying KVL at output side,

$$V - 2I_C - V_{CE} = 0$$

$$V_{CE} = V - 2I_C$$

$$= 9 - 2 \times 0.65$$

$$= 7.7 \text{ V}$$

Operating point = (V_{CE}, I_C)

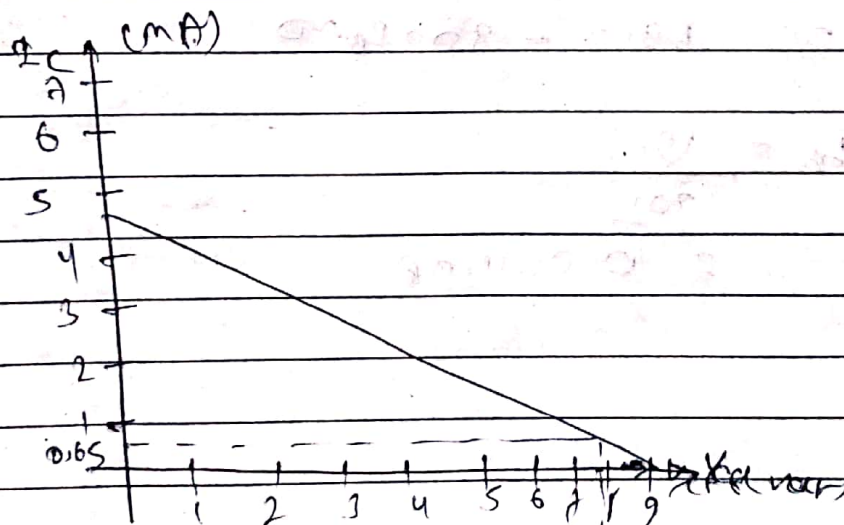
$$= (7.7 \text{ V}, 0.65 \text{ mA})$$

For DC load line,

put $I_C = 0$ in eqn (1), $V_{CE} = 9 \rightarrow (9 \text{ V}, 0 \text{ A})$

put $V_{CE} = 0$ in eqn (1), $I_C = \frac{9}{2} \rightarrow 4.5 \text{ mA}$

$\rightarrow (0, 4.5 \text{ mA})$



$(7.7, 0.65)$

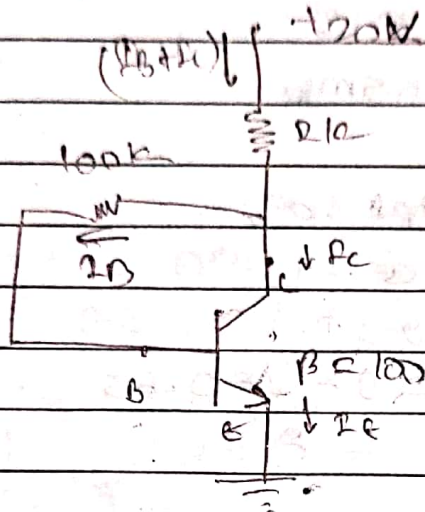
Sketch the CBT shown below, find

i) D.P.

ii) Stability Factor

iii) DC load line and show operating point on it

→



→

Apply KVL at input;

$$20 - 2(I_B + I_C) - 100I_B - V_{BE} = 0$$

$$20 - 2I_B - 2I_C - 100I_B - V_{BE} = 0$$

$$-102I_B - 2I_C - V_{BE} + 20 = 0$$

$$19.3 - 102I_B - 2\beta I_B = 0$$

$$19.3 - 302I_B = 0$$

$$I_B = \frac{19.3}{302}$$

$$= 0.064 \mu A$$

$$I_L = \beta I_B$$

$$= 100 \times 0.064$$

$$= 6.4 \text{ mA}$$

Again, applying KVL at output side,

$$20 - 2(I_B + I_C) - V_{CE} = 0$$

$$20 - 2(0.064 + 6.4) - V_{CE} = 0$$

$$V_{CE} = 7.07 \text{ V}$$

$$OP = (V_{CE}, I_C)$$

$$= (7.07 \text{ V}, 6.4 \text{ mA})$$

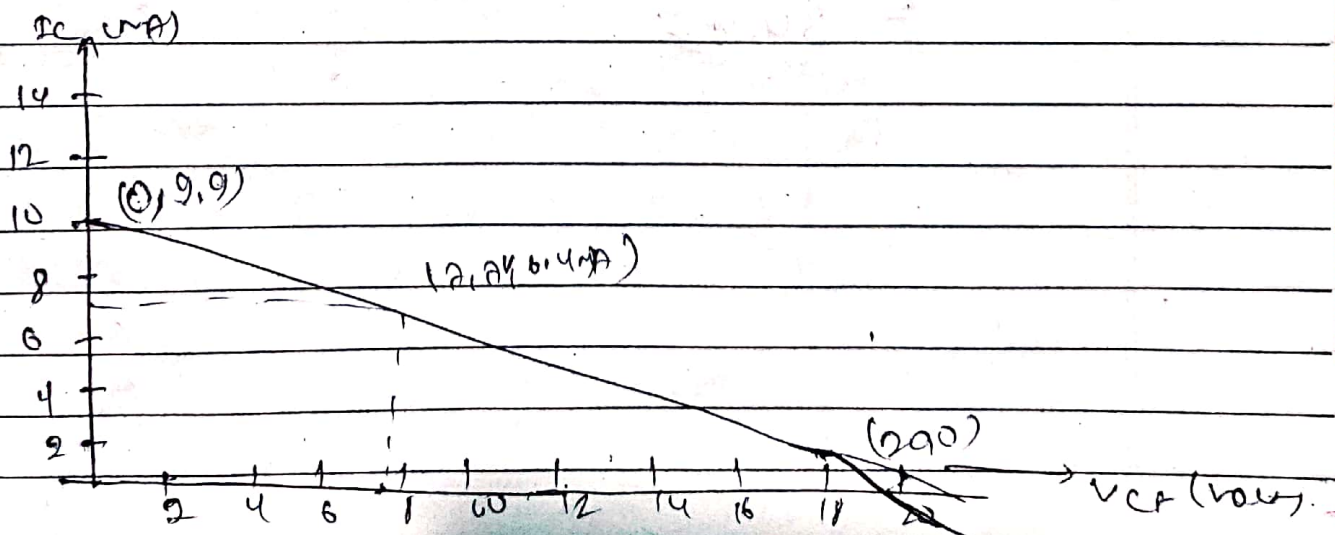
from eqn (1),

$$20 - 2 \left(\frac{I_C}{\beta} + I_C \right) - V_{CE} = 0$$

$$\text{or } 20 - 2I_C \left(1 + \frac{1}{\beta} \right) - V_{CE} = 0$$

$$\text{put, } V_{CE} = 0, \quad I_C = \frac{20}{2(1 + \frac{1}{\beta})} = 9.9 \text{ mA}$$

$$\text{put, } I_C = 0, \quad V_{CE} = 20 \text{ V}$$



Now, we have,

$$S = \frac{\beta + 1}{1 - \beta \cdot \frac{dR_B}{dR_C}}$$

diff w.r to R_C ,

$$\frac{dS}{dR_C} = \frac{dR_B}{dR_C} + \frac{1}{1 - \beta \cdot \frac{dR_B}{dR_C}} \cdot \frac{d}{dR_C} \left(1 - \beta \cdot \frac{dR_B}{dR_C} \right) = 0$$

$$0 = 100 \frac{dR_B}{dR_C} + \frac{2}{1 - 0.96} = 0$$

$$-102 \frac{dR_B}{dR_C} = 2$$

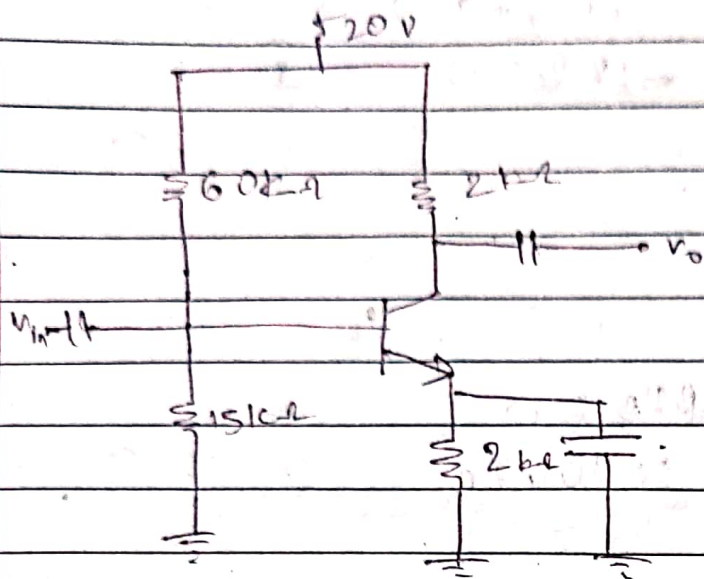
$$\frac{dR_B}{dR_C} = \frac{2}{-102} = -0.0196$$

$$S = \frac{100 + 1}{1 - 100(-0.0196)}$$

$$= 101$$

$$= 34.12$$

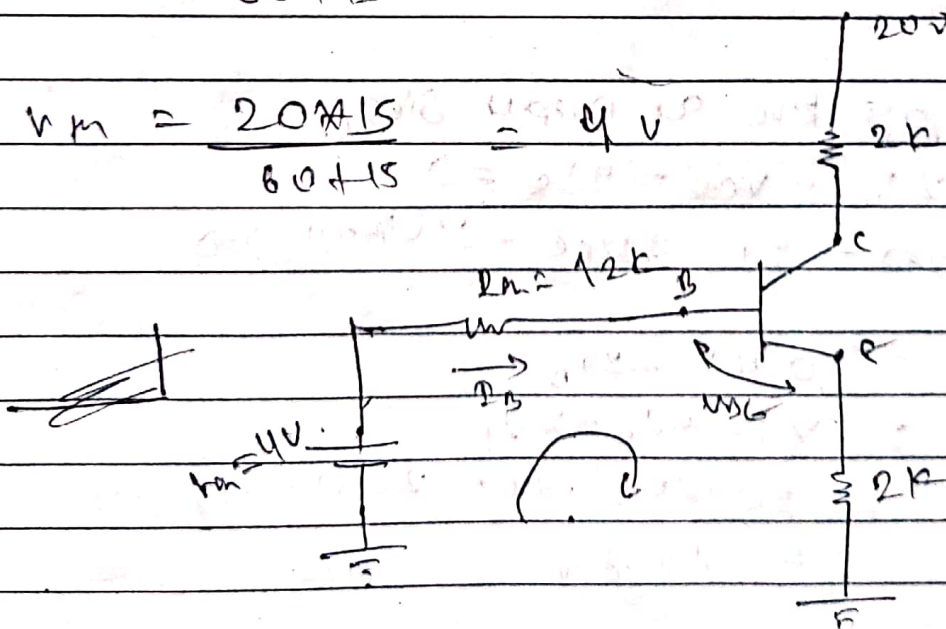
And I_B , I_E , V_C and β for the given
broad circuit from the fig.



Capacitor acts as open circuit for dc analysis.

$$R_m = \frac{60 \times 15}{60 + 15} = 12 \text{ k}\Omega$$

$$V_m = \frac{20 \times 15}{60 + 15} = 4 \text{ V}$$



Applying KVL at input side,

$$V_{BB} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$\Rightarrow 4 - I_B R_B - 0.7 - 2(I_B + I_C) = 0$$

$$3.3 = I_B R_B + 2I_B + 2I_C = 0 \quad \text{--- (1)}$$

$$\text{Since, } I_C = \beta I_B$$

$$3.3 = I_B R_B + 2\beta I_B = 0$$

$$\Rightarrow 3.3 = I_B R_B + 2 \times 100 I_B$$

$$\Rightarrow 214 I_B = 3.3$$

$$I_B = \frac{3.3}{214} = 0.0154 \text{ mA}$$

$$I_C = \beta I_B = 100 \times 0.0154 = 1.54 \text{ mA}$$

Applying KVL at output side,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\Rightarrow 20 - I_C R_C - V_{CE} - 2(I_B + I_C) = 0$$

$$V_{CE} = 20 - I_C R_C - 2I_B$$

$$= 20 - I_C R_C - 2I_B$$

$$= 20 - 1.54 \times 10 - 2 \times 0.0154$$

$$= 13.8 \text{ V}$$

Stability factor,

$$S = \frac{\beta + 1}{1 - \beta \frac{dI_B}{dI_C}}$$

Diff eqn wrt ϵ_c

$$\frac{d(3.3)}{d\epsilon_c} - 14 \frac{dP_B}{d\epsilon_c} - 2 \frac{dP_c}{d\epsilon_c} = 0$$

$$0 - 14 \frac{dP_B}{d\epsilon_c} - 2 = 0$$

$$\frac{dP_B}{d\epsilon_c} = \frac{2}{14} = -\frac{1}{7}$$

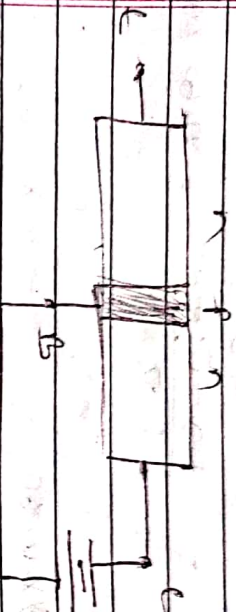
$$S \geq 100 \text{ f1}$$

$$1 - 100 \times \left(-\frac{1}{7}\right)$$

$$= \frac{101}{10} \text{ A}$$

$$= 10.1 \text{ A}$$

Reverse through or punch through.



The reverse voltage across collector base junction increases the depletion width across the collector base junction. Since the base is lightly doped, the depletion region penetrates deeper into the base with this reverse biasing. As a result, the effective base width reduces and consequently the base current is reduced. This effect is called Early effect.

If reverse bias voltage across collector base junction is further increased, the effective base width tends to zero i.e. the depletion width completely covers the base. This effect is termed as punch through or punch through.

In this condition the base doesn't exist and the emitter effectively behaves as a short circuit with the collector and cause of voltage breakdown.

Design of transistor biasing circuit:

The following guidelines may be followed for the designing the transistor biasing circuit,

- ↳ (i) The voltage across R_E is taken as 10% - 20% of V_{CC} .
- (ii) I_E should be chosen in such a way that operating point lies at the middle of the load line.
- (iii) The voltage across the collector-emitter terminals (V_{CE}) should be 50% of V_{CC} .
- (iv) The voltage drop across R_E will be 30-40% of V_{CC} .
- (v) The value of resistance R_1 & R_2 are selected so as to give,

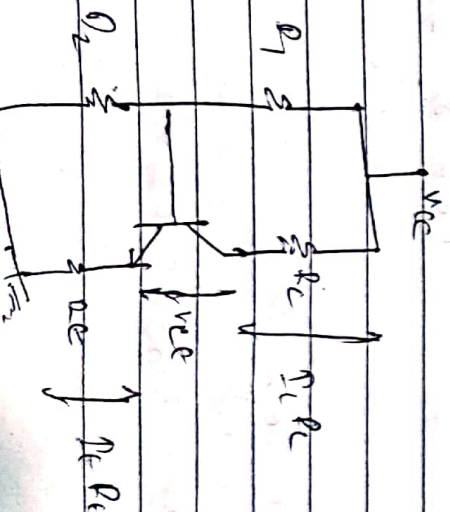
$$I_1 \geq 10 I_B \text{ or } I_1 \geq 10 \frac{I_E}{\beta}$$

$$R_{BE} = 10 R_E$$

where,

$$R_{BE} = R_1 \parallel R_2$$

Q Design a CE (common emitter) self bias circuit for $I_E = 2 \text{ mA}$. Assume $V_{CC} = 10 \text{ V}$, $\beta = 100$.



$$\beta \leq 100$$

$$V_{CE} = 20V$$

$$R_C = 2M\Omega$$

For voltage drop across $R_C = 10V$ of V_{CE}

$$I_C R_C = \frac{20 \times 10}{100}$$

$$R_C = \frac{2}{I_C}$$

$$R_C = 2M\Omega$$

$$I_C = I_E = 2mA$$

$$R_E = \frac{8}{2} = 4\Omega$$

$$V_{CE} = 50\% \text{ of } V_{CC}$$

$$= 20 \times 0.5$$

$$= 10V$$

Voltage drops across $R_C = 40\% \text{ of } V_{CC}$

$$I_C R_C = 40 \times 50$$

$$I_C R_C = 8$$

$$R_E = \frac{8}{2} = 4\Omega$$

$$I_1 = 10\mu A$$

$$= 10V \frac{R_C}{R_E}$$

$$= 10V \frac{2}{100} = 0.2mV$$



$$I_2 = I_1 - I_B = 0.1 - 0.02 = 0.08 \text{ mA}$$

$$V_{CE} - V_{CE} - I_{CE} R_C = 0$$

$$V_{CE} = V_{CE} - I_{CE} R_C = 0.08 \times 1 = 0.08 \text{ V}$$

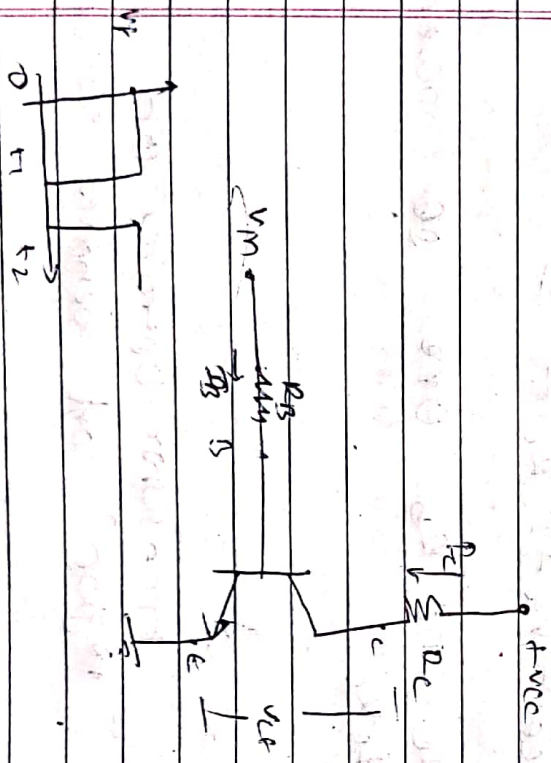
$$V_{CE} = V_{CC} - V_{CE}$$

$$R_1 = \frac{V_{CC} - V_{CE}}{I_1} = \frac{20 - 0.08}{0.02} = 999.2 \text{ k}\Omega$$

$$I_2 R_2 = V_{CE}$$

$$R_2 = \frac{V_{CE}}{I_2} = \frac{0.08}{0.018} = 4.44 \text{ k}\Omega$$

Transistor as a switch:



Let us consider a sat as shown in fig. above, where transistor is operated as a switch, it may operate either in

Saturation region or in cut-off region. when transistor is ON, it operates in saturation region and when transistor is OFF, it operates in cut-off region.

when, $V_{in} = V_F$

$$I_B = \frac{V_F - V_{BE}}{R_B}$$

$$I_B = \frac{V_F - V_{BE}}{R_B}$$

$$I_C = \beta I_B$$

Applying KVL: $V_{CC} - I_C R_C - V_{CE} = 0$ *

$$V_{CE} = V_{CC} - I_C R_C$$

The value of R_C and R_B are so chosen that $V_{CE} = I_C R_C$

$$\therefore V_{CE} = 0. V$$

In this condition, transistor operates at saturation region and behaves as switch ON.

When, $V_{in} = 0$, emitter based junction will no longer forward biased and behaves as switch

$$\therefore I_B = 0 \Rightarrow I_C = \beta I_B = 0$$





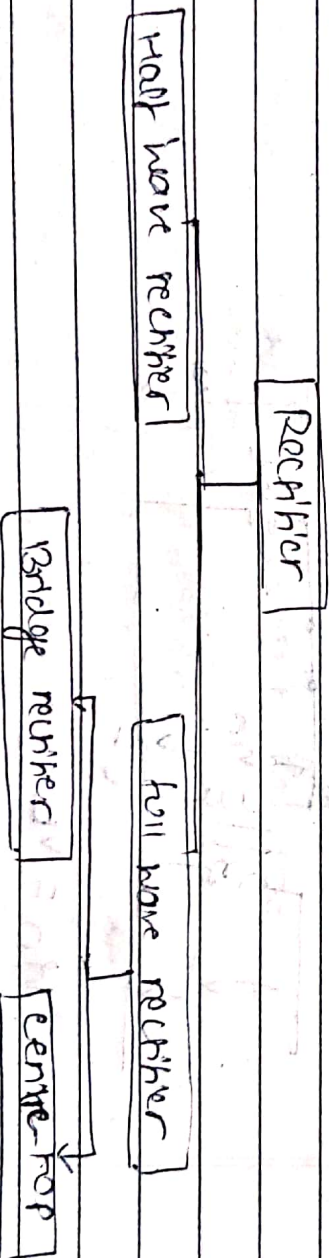
An u_n is u_{on}^{on} , transistor operates at cut-off region and behaves as switch OFF.

CH-3 DC power supply

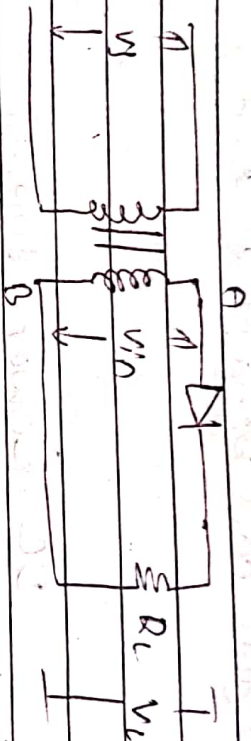
INPUT — Rectifier — filter — voltage regulator — OUTPUT
AC

Rectifier: The process of converting sinusoidal AC voltage into either positive or negative pulsating dc is known as rectification & the corresponding ckt is known as rectifier ckt.

Rectifier ckt are classified as;

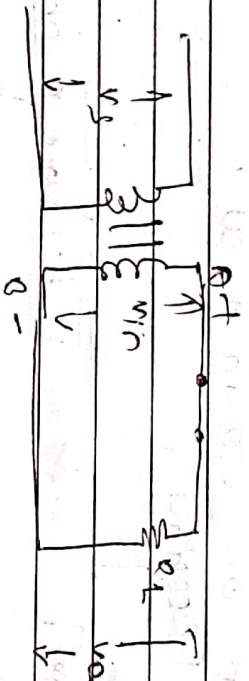


Half wave rectifier :



ckt diagram of half wave rectifier ckt,
 as shown in fig. above. It uses a diode
 which is connected in series with
 load as shown in fig.

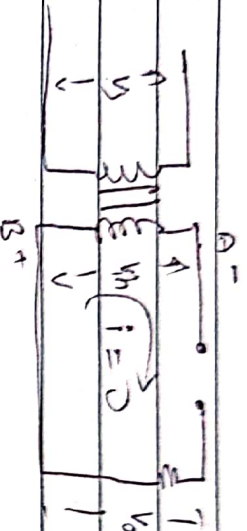
During the half of input, let us
 assume that terminal A is the
 positive, then diode will be forward biased
 and ideally behaves as a short, ~~etc~~
 therefore, output voltage equal to v_{in} .



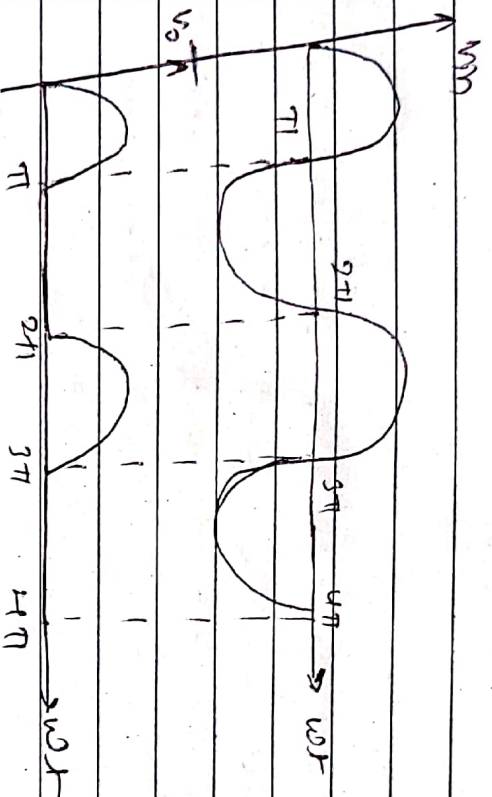
$$v_o = v_i$$

During the half of input, the polarity will
 change and terminal B will be
 positive w.r.t. A, so the diode
 will be reverse-biased, and acts as
 open ckt, so, output voltage $v_o = 0$.

The input and output waveform of half wave rectifier are shown fig below:



$$v_o = 0 \quad \text{for } i = 0$$



i) Average value of output waveform:

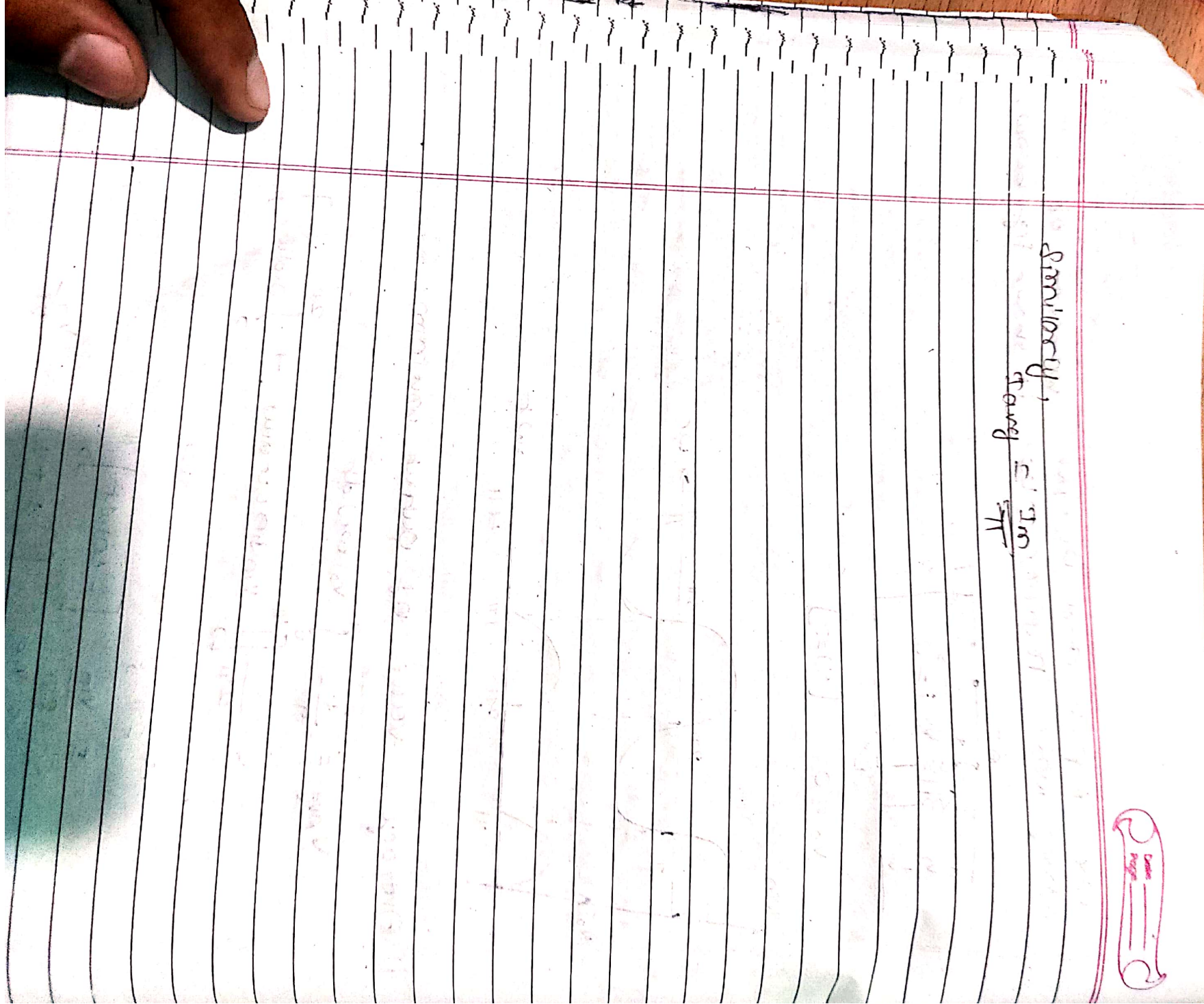
$$V_{avg} = \frac{1}{T} \int_0^T v_o(\omega t) dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} v_m \sin \omega t d\omega t + \int_{\pi}^{2\pi} 0 d\omega t$$

$$= \frac{1}{2\pi} \int_0^{\pi} v_m \sin \omega t d\omega t$$

$$= \frac{v_m}{2\pi} \left[-\cos \omega t \right]_0^{\pi}$$

$$= \frac{v_m}{2\pi} \left[1 + 1 \right] = \frac{v_m}{\pi}$$



11/5/2020
Page 15
Sincerely,

Signature